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School of Bachelors

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGIES ENGINEERING

TOPIC: IMPLEMENTATION OF A REALESTATE MANAGEMENT SYSTEM

OPTION: SOFTWARE ENGINEERING

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GENERAL INTRODUCTION

The rapid evolution of information and communication technologies has transformed many sectors, including real estate. Traditional real estate operations are often characterized by inefficiencies such as poor data management, limited market reach, lack of transparency, and slow communication between stakeholders. To address these challenges, this project focuses on the design and implementation of a Real Estate Management System, a web-based platform that facilitates property listing, management, and communication between administrators, agents, and clients.

This system was developed using Laravel (PHP framework) with a MySQL database, following modern software engineering principles and UML-based system modeling.

STRUCTURE OF THE WORK IN PARTS

This work has three chapters, with chapter two divided into two parts. Details are as follows.

- **Chapter 1**, State of the art of Employee scheduling system. That is, the history, Evolution, Existing Applications to this and conceptualization of the employee scheduling system.
- **Chapter 2** is divided into three parts.
  - **Part, one includes**, Preliminary study(Study of existing system), it's limitations, Proposed solutions, Justification of our chosen system.
  - **Part, two includes,** System requirements (Functional and non-functional requirements).
  - **Part, three includes,** System design(UML diagrams), Needed Resource, Cost estimation.
- **Chapter 3,** Results(application Interface screenshots), Conclusion, References.

CHAPTER 1: STATE OF ART REAL ESTATE MANAGEMENT SYSTEM

1.1 Introduction to Real Estate Management Systems

The real estate sector is one of the oldest and most important sectors of any economy. It plays a critical role in wealth creation, urban development, and social stability by providing housing, commercial spaces, and land for development. Traditionally, real estate operations relied heavily on manual processes, physical documentation, and face-to-face interactions. However, the rapid advancement of Information and Communication Technologies (ICT) has significantly transformed how real estate activities are conducted.

The emergence of Real Estate Management Systems (REMS) represents a major technological shift in the industry. These systems automate the processes of property listing, client management, agent coordination, communication, and administrative control. This chapter presents a comprehensive state-of-the-art review of real estate management systems, tracing their historical evolution, technological foundations, existing solutions, current trends, challenges, and motivations for developing a customized system.

1.2 Historical Background & Evolution Of Real Estate Management Systems.

Real estate management has its roots in the earliest human civilizations. From the moment human beings began to settle permanently, land and property became valuable resources. In ancient societies such as **Mesopotamia, Egypt, Greece, and Rome**, land ownership was directly linked to power, wealth, and social status.

In ancient Mesopotamia (around 3000 BC), clay tablets were used to record land ownership, lease agreements, and property transactions. These records represent the earliest known form of real estate documentation. Similarly, ancient Egyptian civilization maintained land registries for taxation and agricultural management along the Nile River.

In ancient Rome, real estate management became more structured. The Romans introduced legal concepts such as:

- Property ownership rights
- Land leases
- Rental agreement
- Property taxation

Roman law laid the foundation for modern real estate regulations and property management principles that are still in use today.

Table 1: Historical Timeline of Real estate management system

Period	**Time Frame**	**Key Characteristics**	**Tools & Methods Used**	**Limitations**
Ancient Civiliazations	3000 BC - 500 AD	Land ownership linked to power and wealth; early leasing and taxation systems	Clay tablets, papyrus, oral agreements	No standard documentation, limited accessibility
Roman Era	500 BC - 476 AD	Formal property laws, rental agreements, land taxation	Written contracts, legal codes	Manual record keeping, centralized control
Medieval Period	5^{th} - 15^{th} Century	Feudal land ownership; landlords controlled property; peasants paid rent	Church records, oral contracts	Lack of transparency, poor record management
Industrial Revolution	18^{th} - 19^{th} Century	Urbanization increased demand for housing and	Paper files, ledgers, land registries	Slow data processing, high error rates

		commercial property		
Early 20th Century	1900 - 1960	Organized real estate agencies; administrative offices emerged	Filing cabinets, paper forms	Data duplication, risk of loss
Early computer Era	1960 - 1980	Introduction of computers for property record storage	Standalone computers, spreadsheets	High cost, limited access, no networking
Database Systems Era	1980 - 1995	Use of relational databases for structured data management	Local databases (SQL-based)	Required technical expertise, limited scalability
Internet &Web Era	1995 - 2010	Online property listings and global reach	Websites, email communication	Security concerns, limited interactivity
Mordern Digital Systems	2010 - present	Fully integrated platforms with analytic and messaging	Web frameworks (Laravel), cloud services	Requires stable internet and maintenance
Proposed System	Present	Secure, scalable, role-based real estate management	Laravel, MySQL, MVC architecture	Dependent on hosting and technical support

1.3 Existing Systems in Real Estate Management

Today, numerous real estate management platforms are used globally to facilitate property listing, client management, communication, and transaction tracking. These systems aim to digitize traditional real estate operations by allowing agents to manage properties, buyers, sellers, and inquiries through centralized platforms. Common features include property listing, search and filtering, messaging, approval workflows, and reporting dashboards.

Although these platforms provide valuable services, many are either too complex, too expensive, or not well adapted to local real estate markets such as those found in developing countries. Some focus heavily on large agencies, while others lack proper role-based access, approval mechanisms, or performance tracking for agents.

Studying existing real estate management systems helps identify strengths, weaknesses, and gaps, which guided the design of the proposed Web-Based Real Estate Management System developed in this project.

Table 2: Overview Existing Real Estate management system

Application	Key Features	Strengths	Limitations
Zillow	Property listings, search filters, agent profiles, messaging	Large user base, powerful search engine	Not suitable for internal agency management
Realtor.com	Listings, property analytics, agent communication	Accurate market data, trusted platform	Limited customization for agencies
Propertybase	CRM, property listings, analytics, lead management	Enterprise-grade features, strong CRM	High cost, complex setup
Buildium	Property management, tenant tracking, payments	Property management, tenant tracking, payments	High cost, complex setup
AppFolio	Property accounting, leasing, reporting	Strong financial tools	Expensive and complex for small agencies
Zoho Real	Lead tracking, workflow	Integrates with Zoho	Requires

Estate CRM	automation, reporting	ecosystem	customization for full usability
Local Manual Systems	Paper files, spreadsheets, phone calls	Low cost, familiar to users	Data loss risk, poor scalability, inefficiency
Proposed Systems	Property listing, agent dashboard, messaging, approval workflow	Tailored to local needs, role-based access, cost-effective	Requires internet and basic technical maintenance

1.4 Conceptualization of the Real Estate Manager

The conceptualization of a Real Estate Management System involves translating the operational requirements of property management, sales, and client interaction into a structured and efficient software solution. The core idea is to design a centralized digital platform that simplifies property listing, client communication, and transaction management while minimizing manual errors and administrative overhead.

By analyzing the workflows of administrators, agents, and clients, the system is designed to automate repetitive tasks such as property posting, inquiry handling, and approval processes, while still allowing human oversight for decision-making processes like property validation and negotiations.

The Real Estate Management System is conceptualized around the following key principles:

a. Role-Based Access Control

The system implements role-based access control to ensure that different users have access to functionalities appropriate to their responsibilities.

- Administrators manage users, approve properties, and monitor system activity.
- Agents create and manage property listings, respond to client inquiries, and track performance.
- Clients browse properties and communicate with agents.

This approach ensures data security and prevents unauthorized access to sensitive operations such as property approval and user management.

b. Property and Listing Management

Agents can create, update, and delete property listings, including details such as price, location, images, and availability status. Administrators validate and approve listings before they are made publicly visible. This workflow ensures data accuracy and maintains platform credibility while allowing agents to manage their portfolios efficiently.

c. Client Interaction and Messaging

The system enables seamless communication between clients and agents through an integrated messaging module. Clients can send inquiries directly from property listings, and agents receive and respond to messages in real time. This reduces communication delays and improves customer engagement and satisfaction.

d. Data-Driven Decision Support

The system generates reports and dashboards that provide insights into property performance, agent activity, and client engagement. Metrics such as the number of listed properties, approved listings, and received inquiries support managerial decision-making and help optimize business operations.

By integrating these core functionalities, the Real Estate Management System aims to deliver a secure, scalable, and user-friendly platform that addresses the limitations of traditional property management methods. The conceptual design serves as the foundation for the system’s architecture, database design, UML models, and user interfaces, ensuring that the final implementation aligns with real-world operational requirements and user expectations.

CHAPTER 2: SYSTEM ANALYSIS AND DESIGN

This chapter focuses on the analysis and design of the Real Estate manager. It is divided into three parts:

1. **Part One:** Preliminary Study examines the current scheduling systems, their limitations, and justifies the proposed solution.
2. **Part Two:** System Requirements outlines both functional and non-functional requirements of the new system.
3. **Part Three:** System Design presents models, UML diagrams, and resource estimation for the system.

PART ONE: PRELIMINARY STUDY

2.1 Study of Existing Systems

The first step in the development of the Real Estate Management System is the analysis of existing methods used for managing properties, clients, and real estate transactions. Both traditional (manual) and modern (digital) approaches were examined in order to identify their strengths, weaknesses, and operational challenges. This study provides a foundation for proposing an improved and more efficient system.

a. Manual Property Management Methods

In many small and medium-scale real estate agencies, property management is still carried out using manual processes. These methods rely heavily on paperwork and face-to-face communication.

- Property details are recorded in notebooks, files, or printed documents.
- Property listings are advertised using physical posters, banners, or word-of-mouth.
- Client inquiries are handled through phone calls or in-person visits.
- Property approvals and updates are managed informally without a centralized record.

- Tracking agent performance and property status requires manual follow-ups.

2.1.2 Limitations of Manual Methods

- High risk of data loss or damage to physical records.
- Difficulty in updating property information in real time.
- Inefficient communication between agents, administrators, and clients.
- Lack of transparency and accountability.
- Time-consuming administrative processes.

The preliminary study shows that manual methods are inefficient, error-prone, and difficult to manage, while digital solutions are either too general or too costly for smaller organizations. This highlights the need for a specialized Real Estate Management system that is cost-effective, user-friendly, and tailored to the organization's needs.

2.1.3 Proposed Solutions

To address the limitations identified in existing real estate management methods, several possible solutions were examined. The primary objective is to design an efficient, secure, and user-friendly system that simplifies property management, enhances communication between agents and clients, and supports administrators in monitoring and approving activities. Three potential approaches were evaluated as follows:

Solution 1: Improved Manual Real Estate Management

This approach focuses on optimizing traditional paper-based or spreadsheet-based property management methods. Standardized forms and templates can be introduced to record property details, client information, and transaction records. Property listings may be shared through printed brochures, emails, or simple online advertisements.

Advantages:

- Low implementation cost.
- No need for advanced technical infrastructure.
- Familiar to staff with limited technical skills.

Limitations:

- Still highly prone to human error and data inconsistency.
- No real-time updates for property availability or inquiries.
- Inefficient communication between agents, administrators, and clients.
- Difficult to scale as the number of properties and clients increases.

Solution 2: Commercial Real Estate Platforms

This solution involves adopting existing online real estate platforms or commercial property management software. These platforms typically offer features such as online property listings, basic messaging, analytic, and sometimes customer relationship management tools.

Advantages:

- Ready-made and reliable solutions.
- Provide automation for property listings and inquiries.
- Often accessible via web and mobile applications.

Limitations:

- Subscription and service costs can be high, especially for small agencies.
- Limited customization to match specific organizational workflows.
- Some features may be unnecessary or overly complex.
- Dependency on third-party platforms for data and system availability.

Solution 3: Custom Real Estate Management System (Our Project)

The proposed solution is the development of a custom Real Estate Management System tailored specifically to the needs of small to medium-scale real estate agencies. This system integrates essential functionalities such as role-based access control, property listing and approval workflows, client-agent messaging, automated notifications, and performance dashboards.

Administrators can manage users and approve property listings, agents can create and manage properties while interacting with clients, and clients can browse listings and make inquiries seamlessly. By focusing on real operational requirements, the system ensures simplicity, efficiency, and scalability.

Advantages:

- Custom-built to match organizational workflows.
- Cost-effective in the long term.
- Secure role-based access control.
- Improved communication and transparency.
- Supports data-driven decision-making through reports and dashboards.

Limitations:

- Requires initial development, testing, and deployment time.
- Maintenance and updates must be managed internally.

Table 3: Comparison of Proposed Solutions for Real Estate Management

Solution	Key Features	Advantages	Limitations
Manual Real Estate Management	Paper records, spreadsheets, phone/email communication	Low cost, easy to implement, no technical dependency	Prone to human error, data inconsistency, no real-time updates, difficult to scale
Commercial Real Estate Platforms	Online property listings, basic messaging, analytic s dashboards	Feature-rich, automated workflows, web and mobile access	Subscription costs, limited customization, unnecessary features for small agencies
Proposed Real Estate Management System	Role-based access, property listing & approval, client-agent messaging, automated notifications, dashboards	Customization, cost-effective, secure, user-friendly, tailored to organizational workflow	Requires development time, testing, and maintenance

2.1.4 Justification of the Chosen System

The selection of the Real Estate Management System as the preferred solution was guided by a detailed analysis of existing property management approaches, organizational needs, and long-term sustainability. While manual methods and commercial real estate platforms provide certain benefits, they do not fully address the specific requirements of small-to-medium real estate agencies seeking efficiency, flexibility, and cost control.

The key reasons for choosing the proposed Real Estate Management System include the following:

a. Tailored Functionality

The proposed system is specifically designed to address the core operations of real estate agencies. Unlike generic commercial platforms, it focuses on essential features such as property listing and approval, agent management, client inquiries, messaging, and performance dashboards. This targeted approach eliminates unnecessary complexity and ensures that every function directly supports business operations.

b. Cost-Effectiveness

Commercial real estate platforms often require recurring subscription fees and charge extra for advanced features. Developing a custom Real Estate Management System eliminates these recurring costs, making it a financially sustainable solution for small-to-medium agencies. The organization only invests in development and maintenance, resulting in long-term cost savings.

c. User-Friendly Interface

The system emphasizes ease of use through a clean and intuitive interface. Role-based access control ensures that administrators, agents, and clients interact only with features relevant to their roles. This reduces the learning curve, minimizes user errors, and improves overall productivity.

d. Flexibility and Scalability

The proposed system is flexible and can be easily extended to support future business needs. New features such as payment integration, advanced analytic, or mobile access can be added without redesigning the entire system. This scalability ensures that the application remains relevant as the organization grows.

e. Improved Communication and Data Accuracy

Integrated messaging and automated notifications enhance communication between agents and clients, reducing delays and misunderstandings. Property approval workflows and centralized data storage improve data accuracy, reduce duplication, and ensure that only verified listings are published. This leads to improved trust, better customer experience, and efficient administrative control

PART TWO: SYSTEM REQUIREMENTS

2.2 System Requirements

System requirements define what the Real Estate Management System (REMS) must accomplish and the conditions under which it must operate. They form the foundation for system design, development, and evaluation. These requirements were identified based on system analysis, stakeholder needs, and project objectives.

The system requirements are categorized into two main groups:

a. Functional Requirements

These describe the specific services, behaviors, and operations the system must provide. They answer the question:

“What must the system do?”

b. Non-Functional Requirements

These define the quality attributes and constraints of the system, such as security, performance, usability, and reliability. They answer the question:

“How should the system perform?”

The following sections present the detailed functional and non-functional requirements of the Real Estate Management System.

2.2.1 Functional Requirements

The functional requirements describe the core operations that the Real Estate Management System (REMS) is expected to perform. These requirements were derived from the analysis of real estate workflows involving administrators, agents, and clients.

A. User Authentication and Account Management

- The system allows users (Administrator, Agent, Client) to register and log in using valid credentials.
- The system validates user roles and assigns appropriate access permissions upon login.
- Administrators are able to create, edit, deactivate, or delete user accounts.
- Passwords are securely stored using hashing and encryption techniques.
- The system allows users to update their personal profile information.

B. Role-Based Access Control

The system restricts access to features based on user roles:

- Administrator

  Full access to user management, property approval, system monitoring, and report generation.

- Agent

  Access to create and manage property listings, respond to client inquiries, and view performance data.

- Client

  Access to browse properties, view details, and communicate with agents.

The system ensures that sensitive actions such as property approval and user management are restricted to administrators only.

C. Property and Listing Management

- Agents are able to create, edit, and delete property listings.
- Each property includes details such as:
  - ✓ Title
  - ✓ Description
  - ✓ Location
  - ✓ Price
  - ✓ Images
- Availability status
- Administrators must approve property listings before they become publicly visible.
- The system prevents the publication of unapproved or incomplete listings.

D. Property Browsing and Search

- Clients are able to browse available properties.
- The system provides search and filtering options based on:
- Location
- Price range
- Property type
- Availability
- Property details are displayed clearly with images and agent contact information.

E. Client Inquiry and Messaging System

- Clients are able to send inquiries directly from property listings.
- Agents receive and respond to inquiries through the system.
- The system maintains a message history between clients and agents.
- Messaging improves communication efficiency and customer engagement.

F. Property Approval Workflow

- The system allows administrators to review submitted property listings.
- Administrators can approve or reject listings.
- Rejected listings may include feedback for agents to make corrections.
- Only approved listings are displayed to clients.

G. Notifications and Alerts

- The system sends automated notifications for key events such as:
- Property approval or rejection
- New client inquiries
- Message replies
- Profile updates
- Notifications ensure transparency and timely communication among users.

H. Reporting and Dashboard Management

- The system generates reports such as:
- Number of listed and approved properties
- Agent activity and performance
- Client inquiries statistics
- Administrators have access to global system reports.
- Agents have access to their individual performance summaries.

I. Logging and Audit Trail

- The system logs important activities such as:
- User registration and deactivation
- Property creation and approval
- Listing updates
- Audit logs are accessible to administrators for monitoring and accountability

2.2.2 Non-Functional Requirements

The non-functional requirements define the quality standards and technical constraints that guide the development and deployment of the Real Estate Management System.

A. Performance Requirements

- The system must load pages and process requests within a reasonable response time.
- Property searches and listing retrieval should execute efficiently.
- The system must support multiple concurrent users without performance degradation.

B. Security Requirements

- User authentication must use encrypted passwords.
- Role-based access control must prevent unauthorized actions.
- Sensitive operations such as property approval and user management must be restricted and logged.
- Input validation must be implemented to prevent SQL injection and other attacks.

C. Reliability and Availability

- The system must operate reliably with minimal downtime.
- Error-handling mechanisms must prevent system crashes.
- Regular backups should be implemented to protect property and user data.

D. Usability Requirements

- The user interface must be intuitive and easy to navigate.
- Clear forms, buttons, and layouts must be used to reduce user errors.
- Users should be able to perform tasks without requiring extensive training.

E. Scalability Requirements

- The system architecture must support future growth in:
- Number of users
- Property listings
- Client inquiries
- Additional features such as online payments or mobile apps can be integrated in the future.

F. Maintainability Requirements

- The system must be modular and well-structured to allow easy updates.
- Code documentation and comments must be provided for future maintenance.
- New modules can be added with minimal impact on existing functionality.

G. Compatibility Requirements

- The system must function across major web browsers (Chrome, Firefox, Edge).
- The interface must be responsive and accessible on desktops, laptops, and mobile devices.

H. Data Integrity Requirements

- All user and property data must be validated before storage.
- The system must prevent data duplication and inconsistencies.
- Audit logs ensure that changes can be traced to responsible users.

PART THREE: SYSTEM DESIGN & COST ESTIMATION

2.3 System Design(UML Diagrams)

2.3.1 Use Case Diagram

The Use Case Diagram below presents an overview of how users interact with the Real Estate Management System. It illustrates the major actors involved in the system, namely the Administrator (Manager), Agent, and Client (User), and the various functions each actor performs within the completed application.

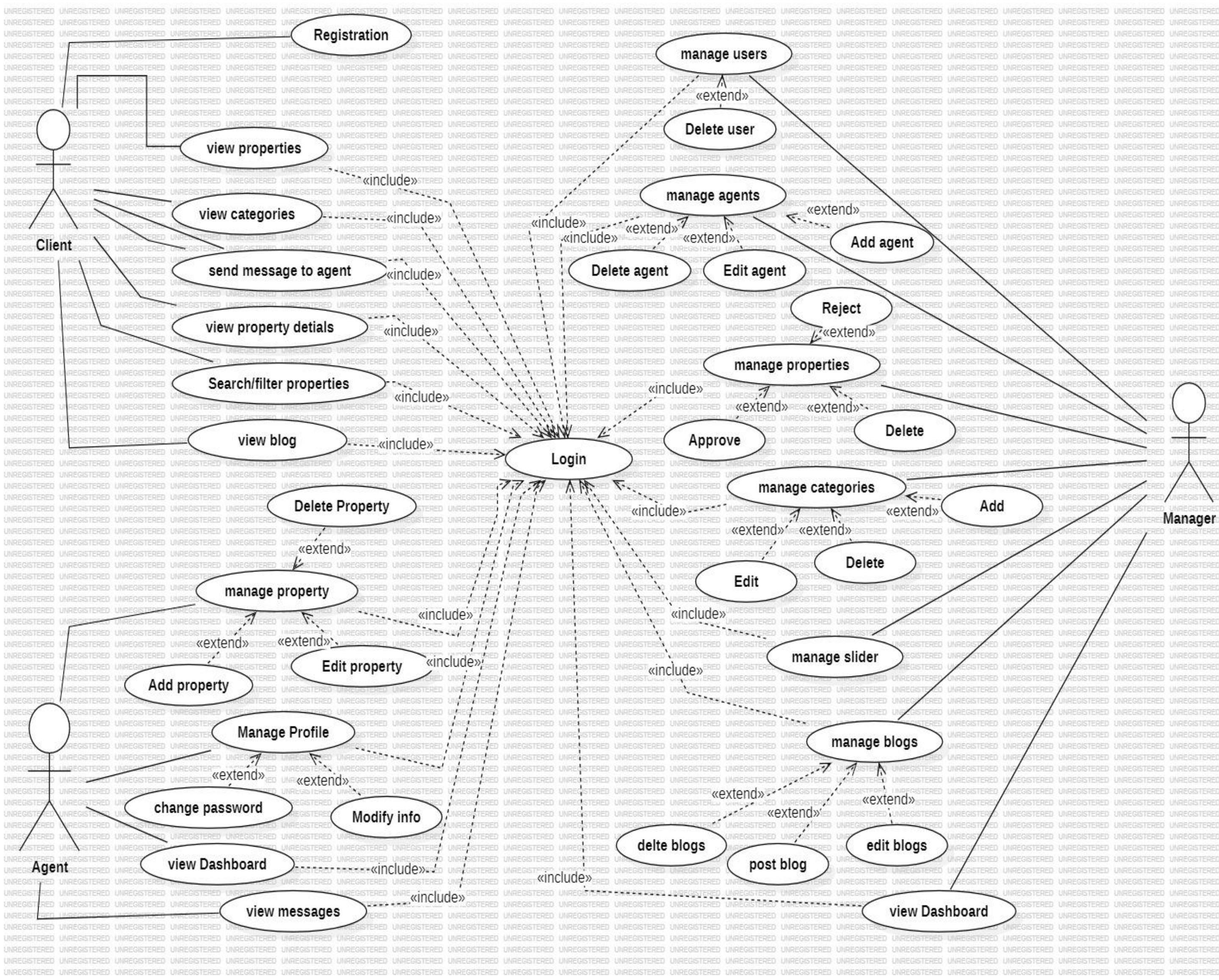

Figure 1 Use Case Diagram

2.3.2 Sequence Diagram

The Sequence Diagram illustrates the interaction flow that occurs when an Agent creates a new property listing in the Real Estate Management System. It presents the step-by-step communication between the Agent, the System Interface, the Property Management Module, the Database, and the Notification Service.

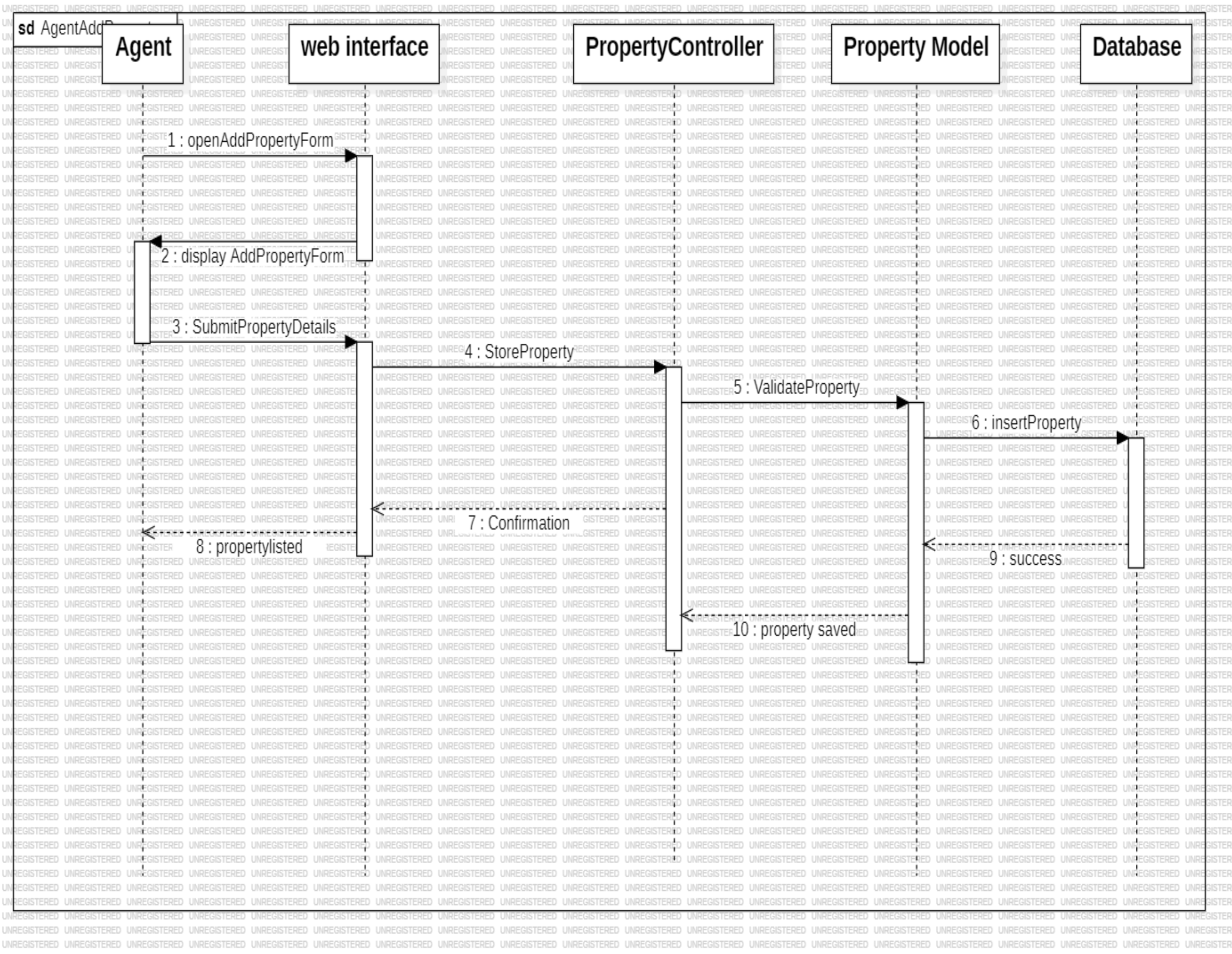

Figure 2 diagram of Add Property by Agent

The Sequence Diagram here illustrates the interaction flow that occurs when an Admin approve a new created property listing in the Real Estate Management System. It presents the step-by-step communication between the Admin, the System Interface, the Property Management Module, the Database.

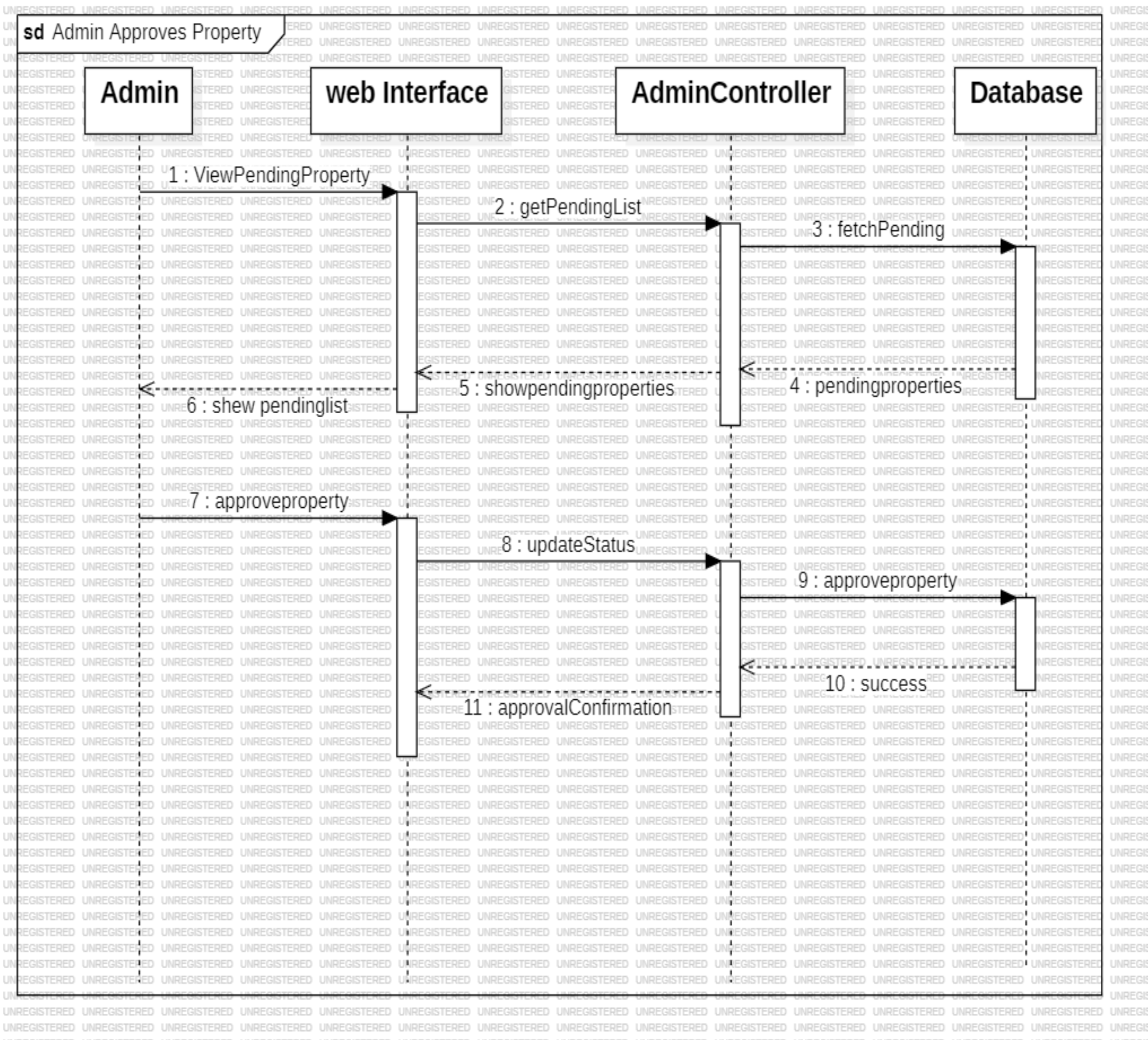

Figure 3: Sequence diagram on Approve Property by Admin

2.3.3 Activity Diagram

The Activity Diagram illustrates the property listing and management process within the Real Estate Management System using three swim lanes: Administrator/Manager, System, and Agent. The diagram provides a clear representation of how activities flow across different actors and how responsibilities are distributed throughout the process.

In the Agent swim lane, the process begins when an agent logs into the system and initiates a request to create a property. The agent enters relevant property details such as location, price, property type, images, and availability status (sale or rent). Once completed, the agent submits the request for approvals

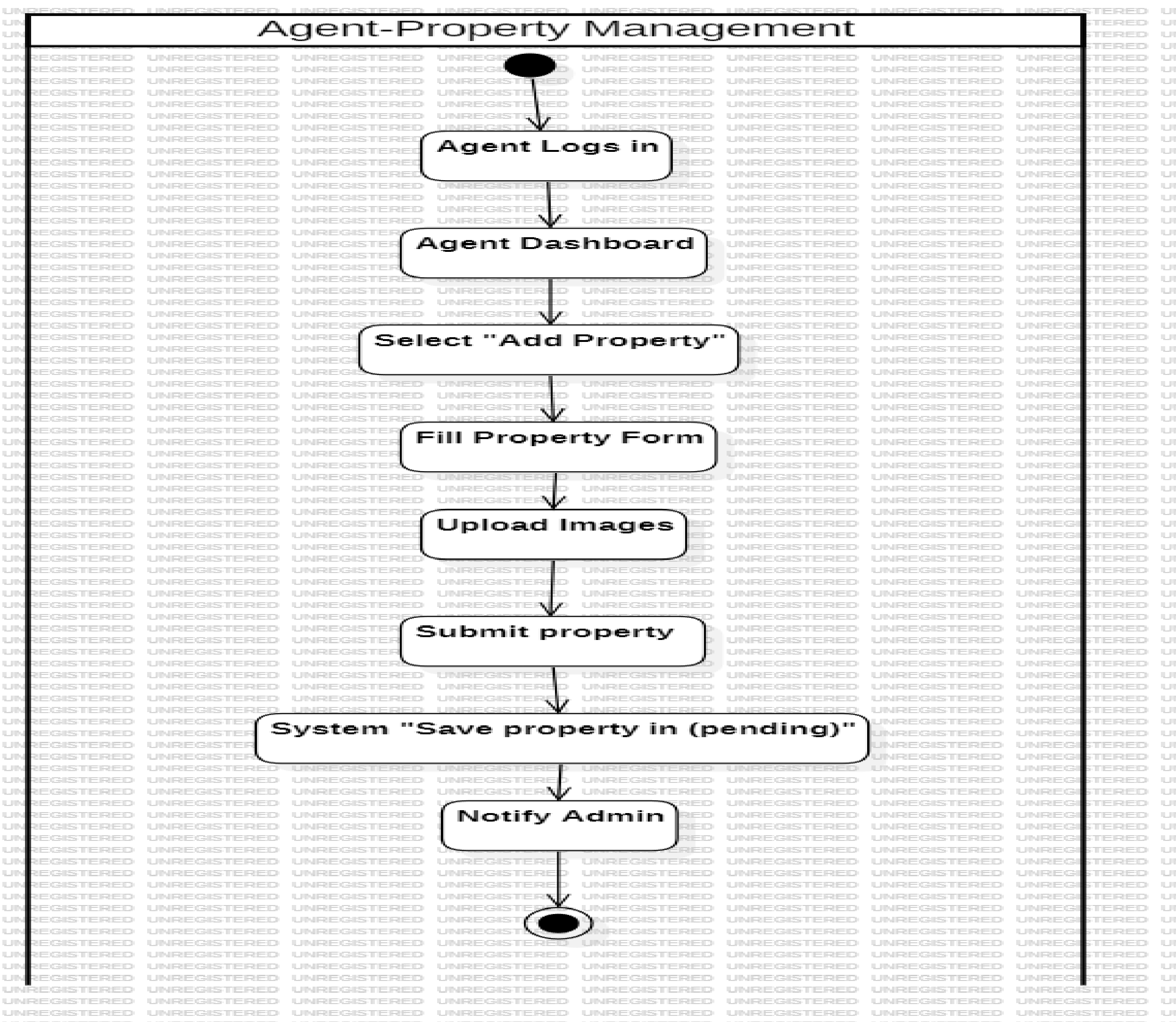

Figure 4Activity Diagram Add property by Agent

In the Agent swim lane, the process begins when an agent logs into the system and initiates a request to create or update a property listing. The agent enters relevant property details such as location, price, property type, images, and availability status (sale or rent). Once completed, the agent submits the request for approval

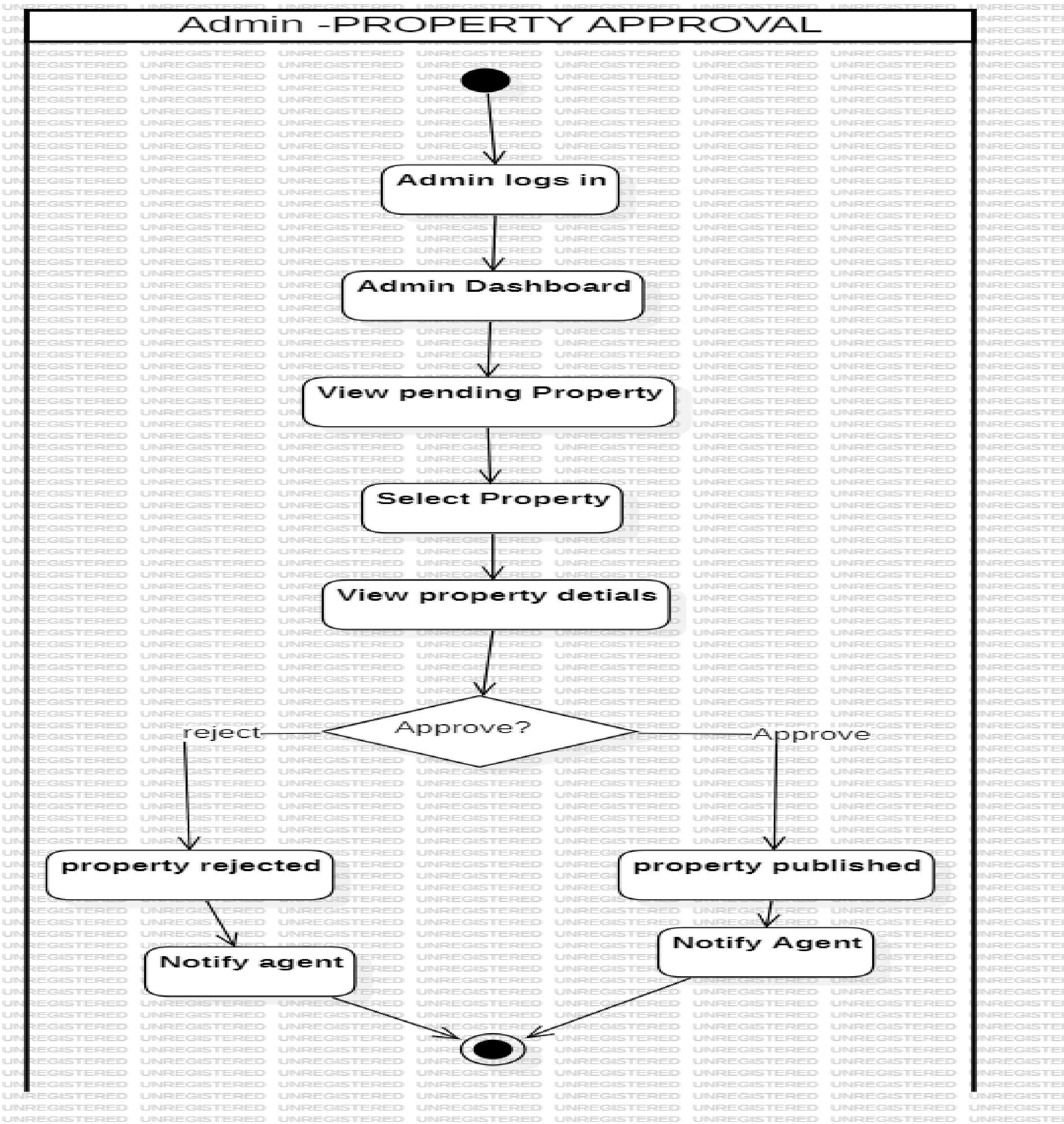

Figure 5 Activity Diagram approve Property by admin

2.3.4 Class Diagram

Class diagram showing the main system classes(entities), their attributes, and relationships within the Real Estate Management System.

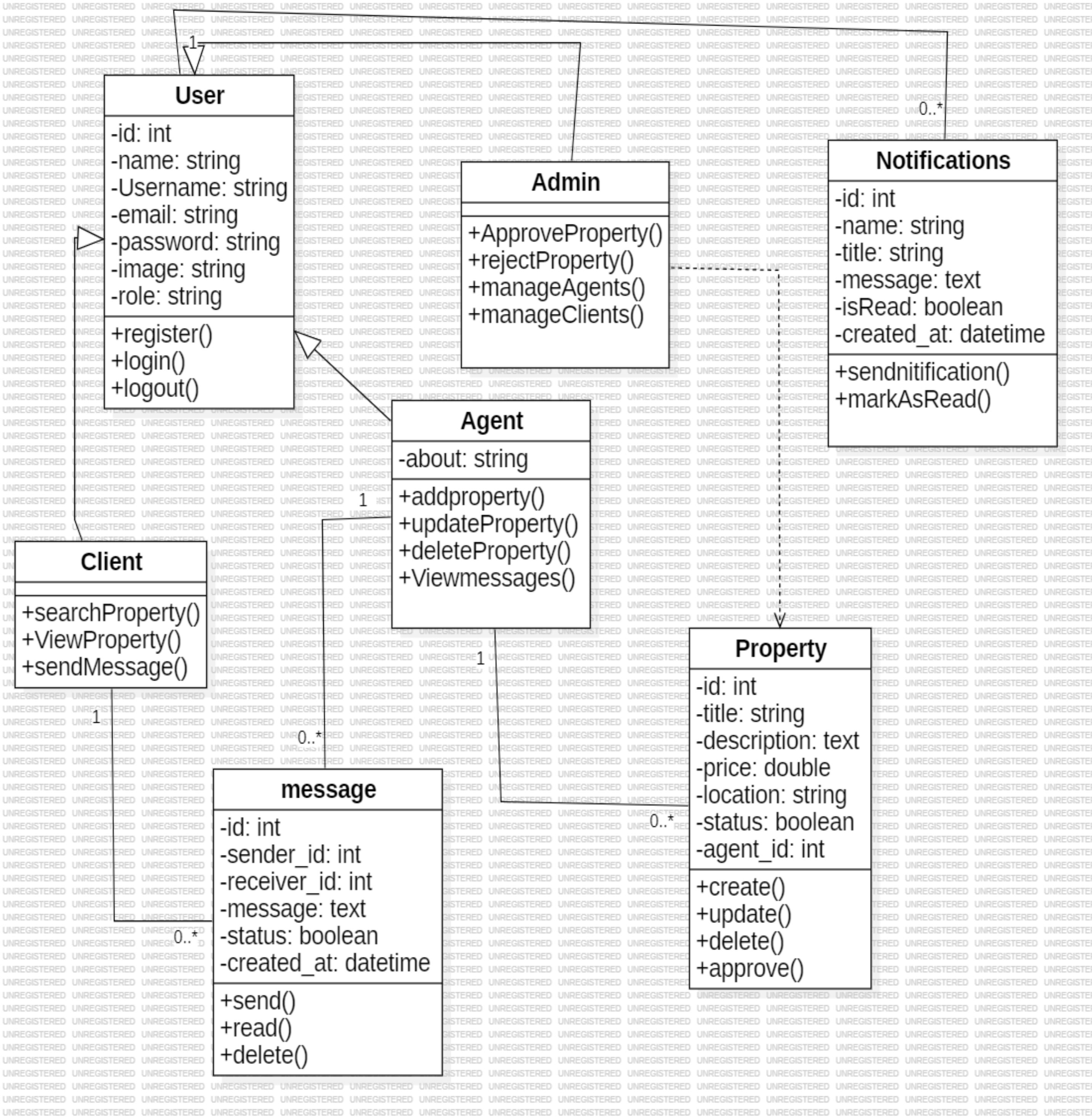

Figure 6 class diagram

2.4 Needed Resources, Cost Estimation

The successful development and deployment of the Real Estate Management System required the availability of adequate resources. These resources are categorized into human resources, hardware resources, and software resources. Proper planning and allocation of these resources ensured efficient system design, implementation, testing, and documentation.

2.4.1 Needed Resources

❖ **Human Resources**

Human resources represent the backbone of the system development process. Each role contributed specific expertise to ensure the system met both functional and non-functional requirements.

- Graphic Designer (01):

  Responsible for creating visual elements such as the system logo, banners, icons, and other graphical assets used in the real estate platform. These visuals enhance the system's appearance and user engagement.

- **System Analyst (01):**

  Responsible for analyzing the project requirements and transforming them into detailed system specifications. The analyst defined functional and non-functional requirements and ensured alignment with user needs.

- **UI/UX Designer (01):**

  Designed the user interfaces and user experience workflows of the system. This role involved modeling system diagrams and creating intuitive layouts for administrators, agents, and clients.

- **Front-End Developer (01):**

  Responsible for implementing the visual components of the system that users interact with, including dashboards, property listings, forms, and navigation elements.

- **Back-End Developer (01):**

  Responsible for server-side logic, database integration, authentication, property management modules, notifications, and role-based access control.

- **Tester (01):**

  Ensured the quality and reliability of the system by conducting manual and automated testing. The tester verified system functionality, security, and performance before deployment.

- **Project Manager (01):**
  Coordinated the project activities, assigned tasks to team members, monitored progress, and ensured that project milestones were met within the specified timeframe.

❖ **Hardware Resources**

The hardware requirements were as follows:

- **A Computer,** having the following characteristics.

Table 4: Characteristics of computer used

Processor	AMD Ryzen 7PRO 2700U w/ Radeon Vega Mobile Gfx (8CPUs), ~2.2GHz
Installed RAM	16.0 GB (15.8 GB usable)
Operating system:	Windows 11 Pro 64-bits
Memory	128MB

- **An external hard drive** to store the backup of our application.
- **A modem** to provide internet connection for research.

❖ **Software Resources**

The software's used for the development of this application are as follows.

◆ Laravel Framework:
Laravel is a PHP-based web application framework that follows the Model–View–Controller (MVC) architectural pattern. It was used to implement the core logic of the Real Estate Management System, including user authentication, role management (admin, agent, client), property listing workflows, messaging, activity logging, and notification services.

◆ **Microsoft Office:** Such as Word, PowerPoint for typing the internship report and designing the power point.

◆ **StarUML:** It is a modelling application that fully supports UML. It was used to design the system blueprints for the use case, class, activity, state, and sequence diagrams.

- **Visual Studio Code:** Best IDE to build rich, beautiful, web platform applications. Used for the development of the project.
- **XAMPP/MYSQL:** Used to implement the database design of the website.

2.4.2 Cost Estimation

The cost estimation is the prediction of cost and prices of resources required by the scope of our project.

- **Human Resources**

Table 5: Cost of Human Resources

Designation	Price per day	Number of days	Total(cfa)
Project Manager	**3000**	**7**	**21,000**
Graphist	**1000**	**7**	**7,000**
Analyst and System Designer	**3000**	**10**	**30,000**
Front-end	**5500**	**15**	**82,500**
Back-end	**6500**	**13**	**84,500**
Tester	**3500**	**12**	**42,000**
Total 1			**267,000**

- **Hardware Resources**

Table 6: Cost of Hardware Resources

Designation	Price(cfa)	Quantity	Total(cfa)
Laptop	**400,000**	**01**	**400,000**
RAM 16GB	**35000**	**01**	**35000**
Hard Disk Drive 500GB to 1terabyte	**30000**	**2**	**60000**
Modem with internet connection	**60000**	**01**	**60000**

Total 2	**580,000**

- **Software Resources**

Table 2: Cost of Software Resources

Designation	Price (cfa)/month	Duration	Total(cfa)
Microsoft Office	**5965**	**3 weeks**	**11950**
StarUMl	**Pending**	**-**	**-**
Visual studio code	**Free**	**-**	**-**
Xampp/PhpMyAdmin	**Free**	**-**	**-**
Total 3			**11950**

- **Summarized Cost Estimation of above Resources**

Table 3: Total Cost Estimation

Total 1	**267000cfa**
Total 2	**580000cfa**
Total 3	**11950cfa**
SumTotal	**858,650frs**

CHAPTER 3: RESULTS AND DISCUSSIONS

This chapter presents the practical outcomes of the Real Estate Management System It includes screenshots of the major application interfaces and discusses how each implemented feature aligns with the system objectives defined in earlier chapters.

3.1 Application Interface Screenshots

3.1.1 Login Page

This interface allows both Client(user), Agent and Admin to securely authenticate into the system. It includes fields for email and password, as well as validation and access control.

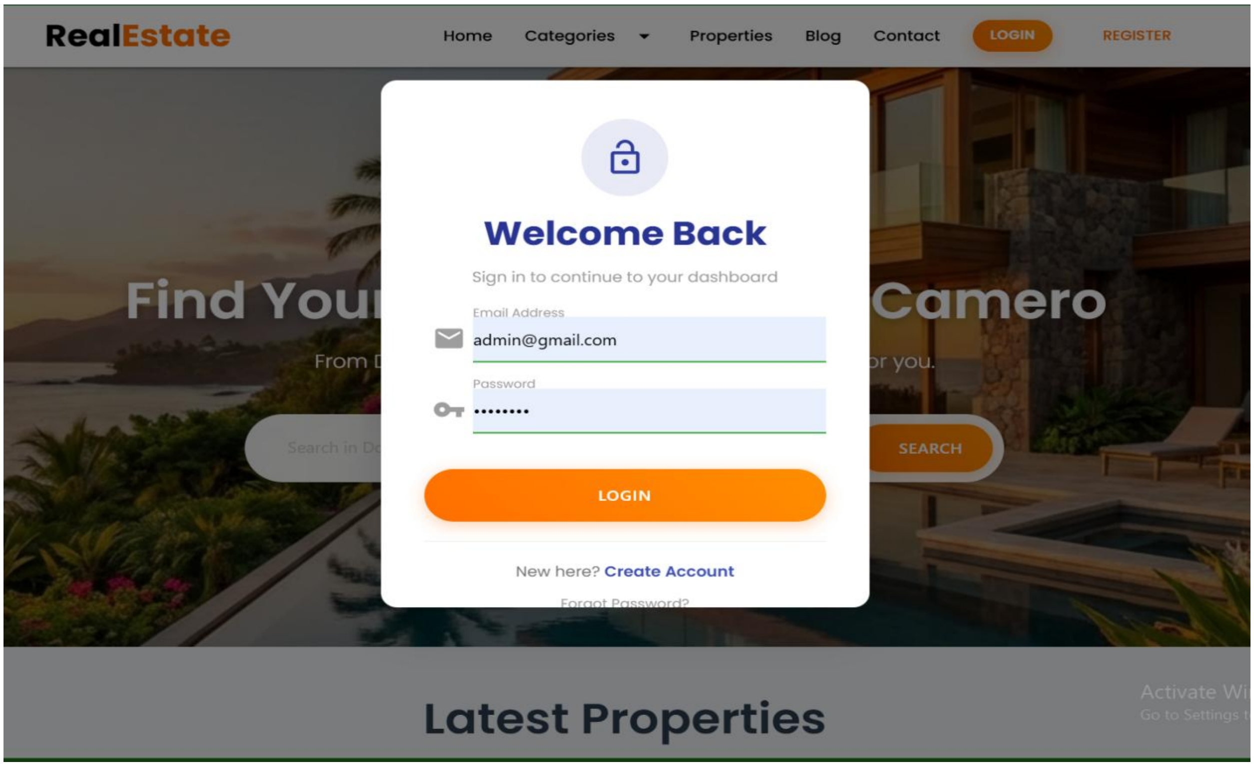

Figure 7: Login Interface

3.1.2 Admin Dashboard

Displays an overview of total e users, total Agent, total properties, and quick-access navigation to core features.

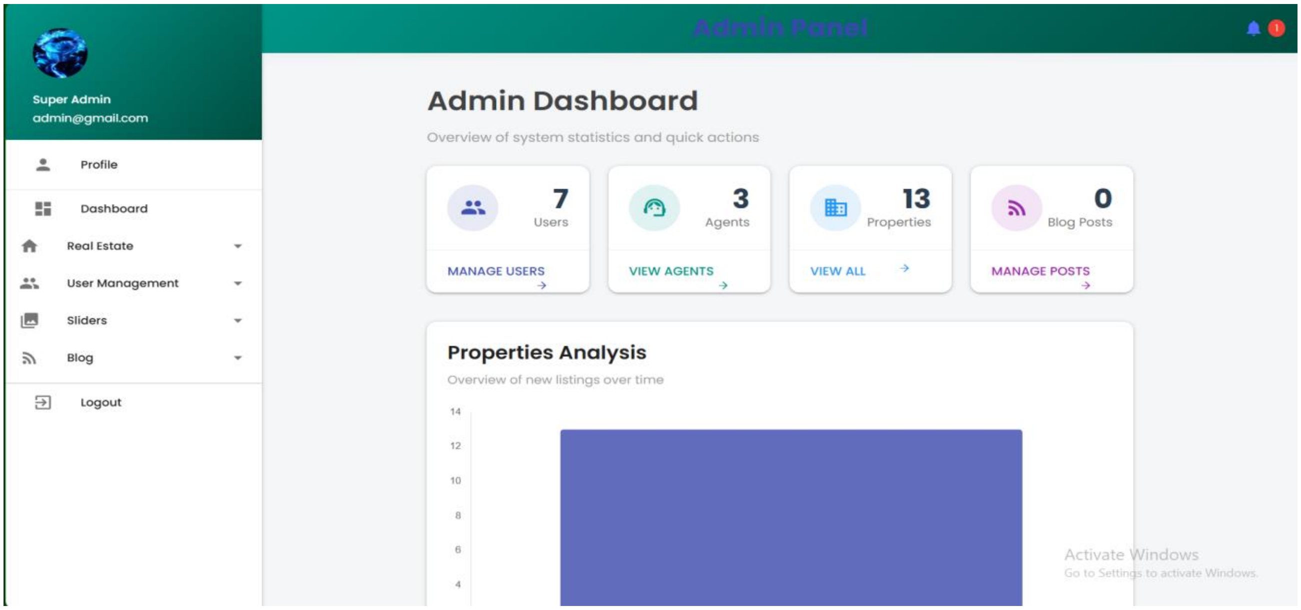

Figure 8: Administrator Dashboard

3.1.3 Create Property Interface

This page enables Agent to create new property, title, price, type, category, location, description

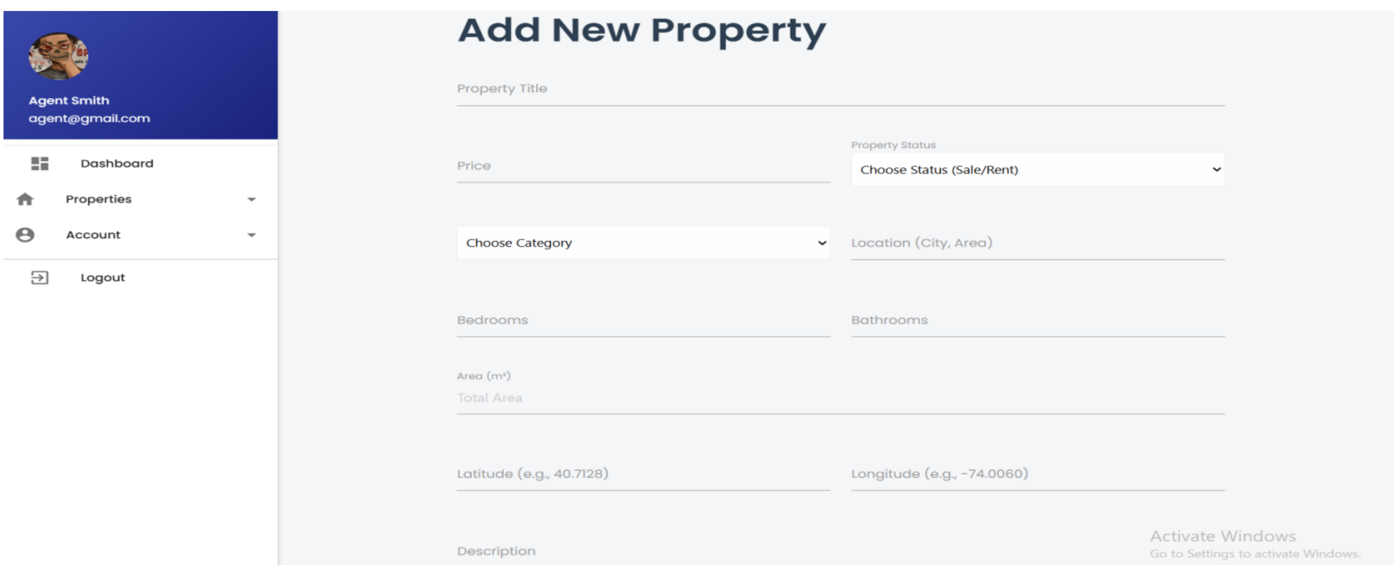

Figure 9: Creating New Property

3.1.4 Agent Dashboard

Shows the agent dashboard shows the total number of all activities carry by the agent like the number of properties added.

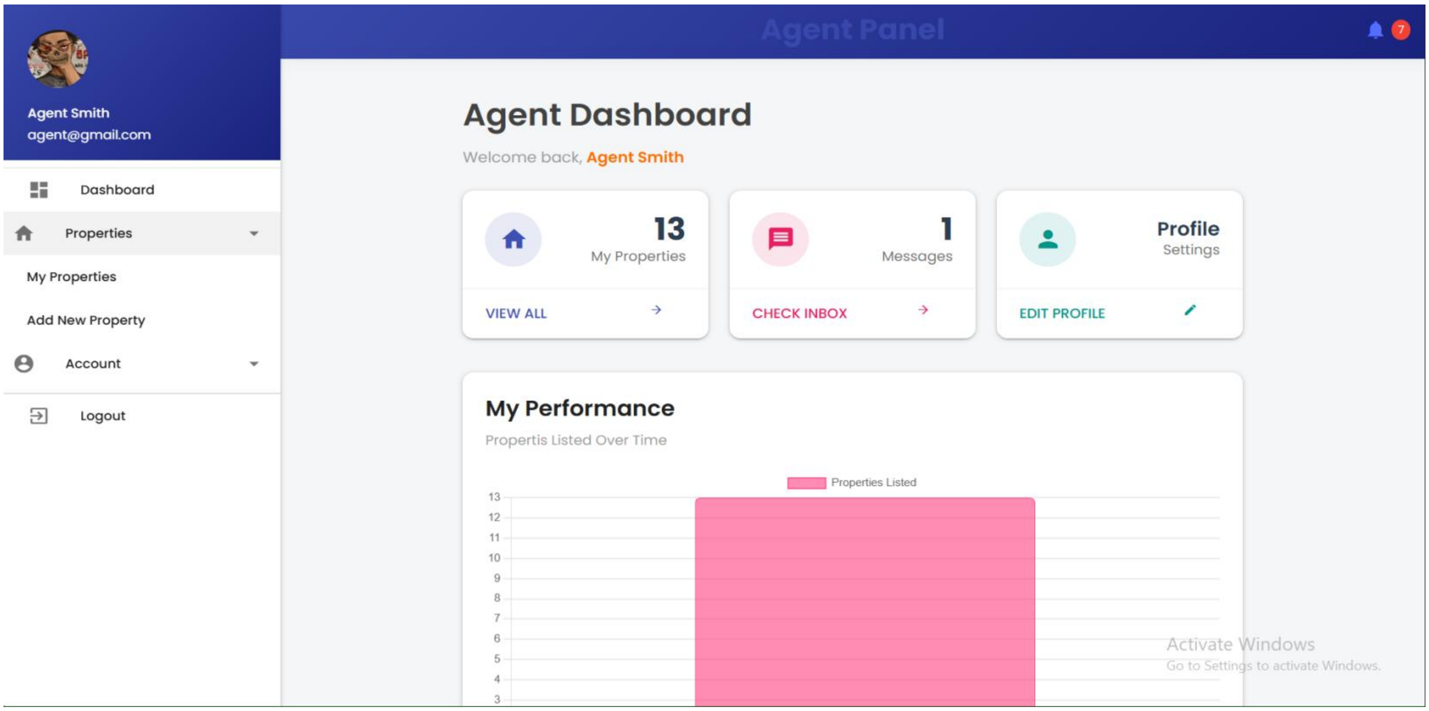

Figure 10: Agent Dashboard

GENERAL CONCLUSION

The Real Estate Management System developed in this project successfully addresses the major challenges associated with traditional property management practices, which are often manual, fragmented, and prone to errors. By digitizing and automating key operations such as property listing, approval workflows, client inquiries, sales and rental tracking, agent performance monitoring, and administrative control, the system significantly improves efficiency, transparency, and operational accuracy.

The application provides a secure and user-friendly platform that supports role-based access for administrators, agents, and clients. Administrators are empowered to manage users, monitor agent progress, approve or deactivate property listings, and receive real-time notifications when properties are sold or rented. Agents can efficiently manage property portfolios, communicate with clients, and record transactions, while clients benefit from easy property browsing and direct interaction with agents.

Through the use of a robust database structure and modern web technologies, the system ensures reliable data storage, real-time updates, and accurate record keeping of all activities within the platform. The integration of notifications, activity logs, and reporting tools enhances accountability and supports data-driven decision-making for management.

Overall, the project successfully achieved its objectives by delivering a fully functional, scalable, and reliable Real Estate Management System tailored to the needs of modern real estate operations. The system demonstrates how technology-driven solutions can transform property management processes, improve service delivery, and support sustainable growth in the real estate sector.

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